

Results and Analysis

The section analyzes the results of the survey we conducted and a statistical analysis of the findings. We aim to answer five research questions: whether users know about AI-related risks, whether they feel at risk, whether awareness affects behavior, which groups appear most susceptible to misconceptions, and whether a short educational intervention improves safer behavior.

A. RQ1: “Do People Know About These Risks?” (Awareness)

To begin, we analyzed whether users are aware of the risks of AI-related privacy and security risks, including their understanding of how AI systems store data, use submitted information, and generate responses.

To do so, we asked respondents several questions regarding these matters. Fig. 1 presents the proportion of correct and incorrect responses that we collected.

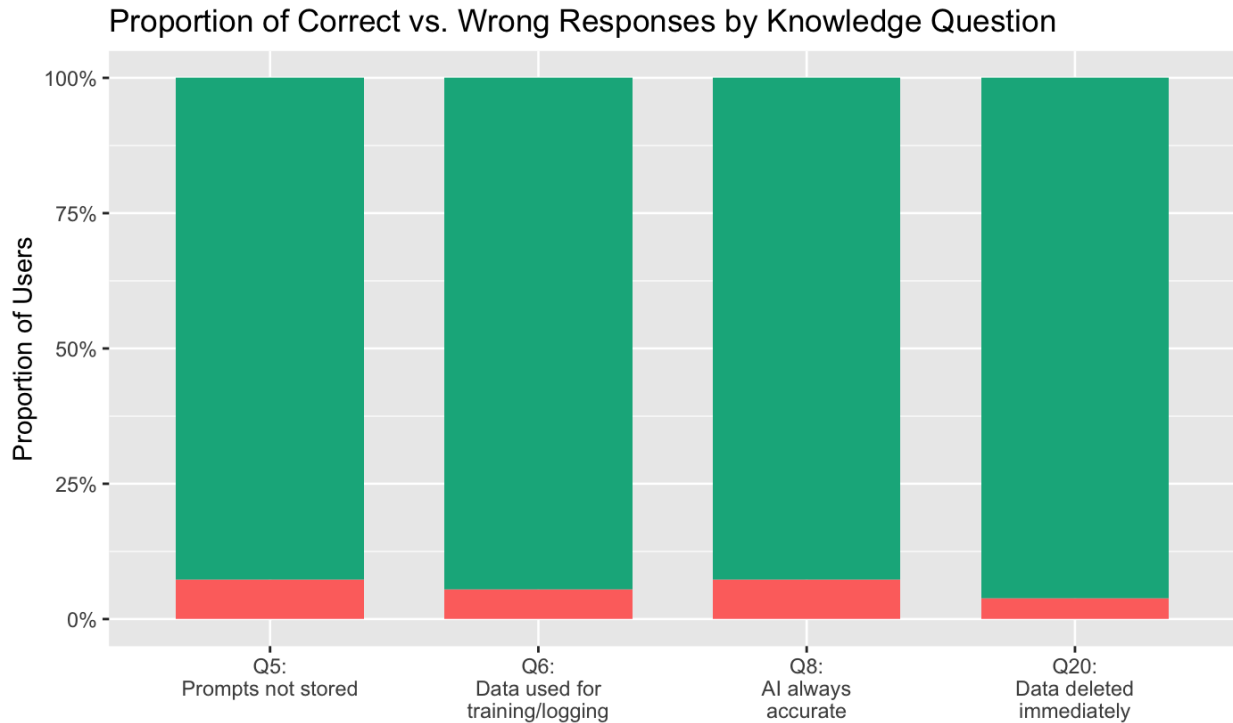


Fig. 1: Count of users who answered incorrectly for knowledge questions.

The data suggest that users are mostly aware of the risks of AI, but a few may lack knowledge of AI's accuracy and data handling. To further our investigation, we assigned a “misconception score” to users and tracked whether the same users answered multiple knowledge questions incorrectly.

Distribution of Misconception Scores

Misconception Score	Count	Proportion
0	45	73.8%
1	5	8.2%
2	4	6.6%
3	7	11.5%
4	0	0%

Table I: Summarization of the amount of misconceptions of respondents throughout the knowledge-testing questions.

In our sample, 73.8% of respondents answered all questions correctly. From the few respondents who incorrectly answered questions, many answered one or two questions

incorrectly. Because of this, we can infer that the majority of users have at least some knowledge of security regarding AI.

To further our investigation, respondents were asked two questions: "How confident are you that you understand where your data goes after submitting it to an AI assistant?" and "How risky do you believe the following scenario is? Uploading tax-related documents to an AI assistant for analysis." The results are visualized in Fig. 2 and Fig. 3.

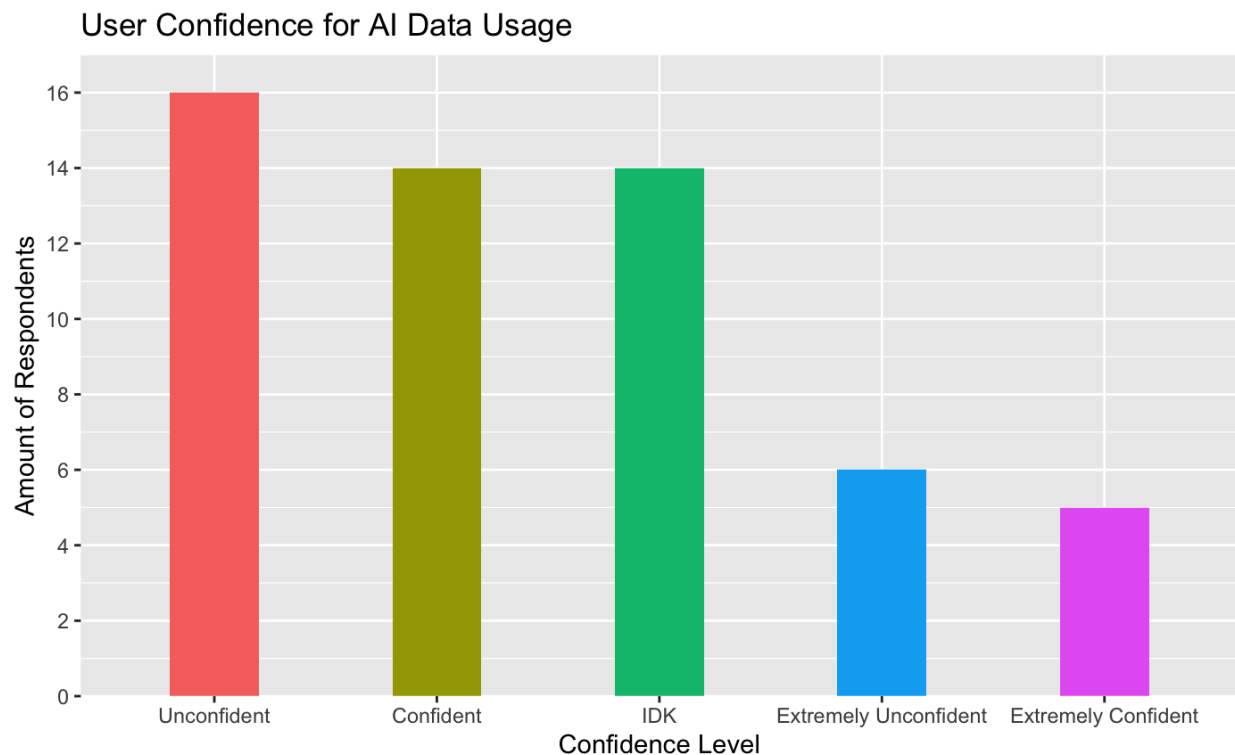


Fig. 2: Distribution of users' responses to how confident they are that they know where their data is going after submitting it to AI.

Although 73.8% of the respondents answered the knowledge-based questions correctly, the large majority of respondents reported that they do not exactly know how AI is handling their data. From this, we can see that users have some perception of risk towards AI; however, they may not have a deep understanding of its inner workings.

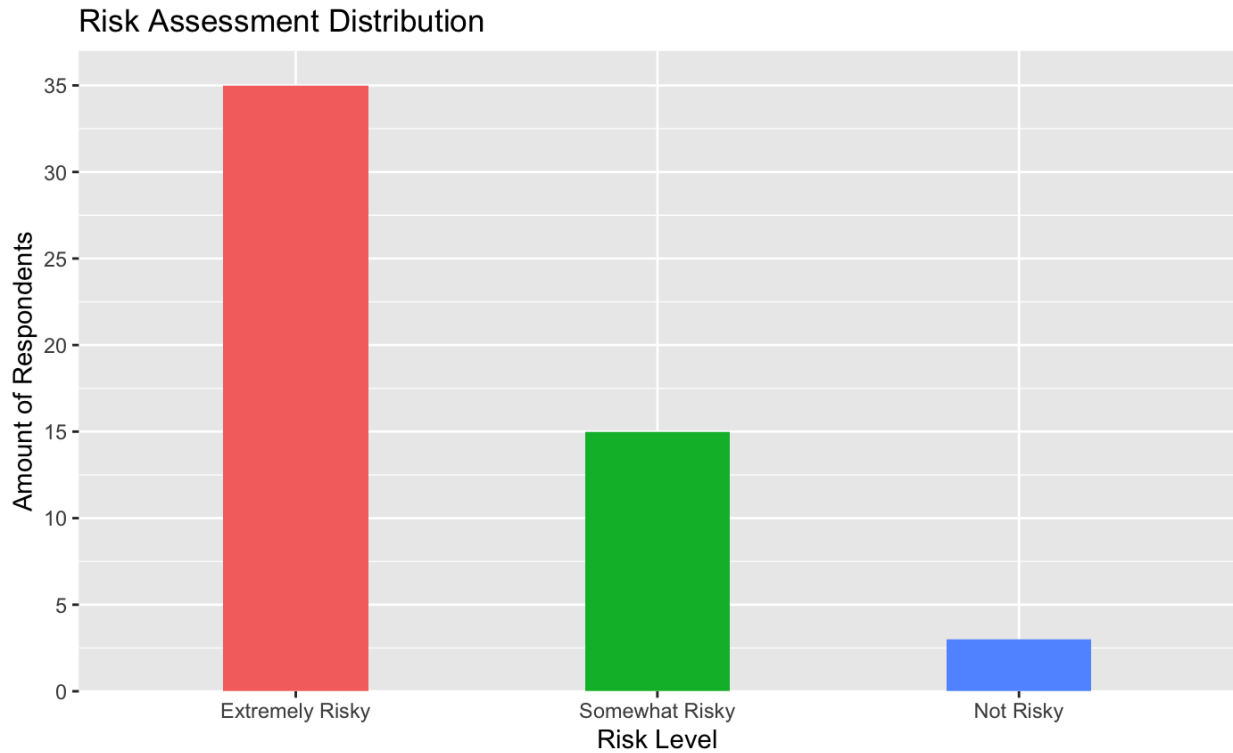


Fig. 3: Distribution of users' risk perception towards uploading tax information into AI.

From the data, we can see that, again, the majority of respondents were able to perceive the tax-information scenario as risky. It further proves that users have a strong surface-level baseline of risks associated with AI, despite possibly not knowing exactly what the risks themselves are.

B. RQ2: “Do Users Feel at Risk?” (Risk Perception)

This subsection examines respondents' trust in AI systems and their perceptions of privacy-related risk when using AI with sensitive information.

Fig. 4 describes the surveyors' trust in AI. In our survey, we found that about 78% of users had little to no trust in AI. Most of the data was clustered on the lower end of trust, while only three users claimed to trust AI a lot and completely.

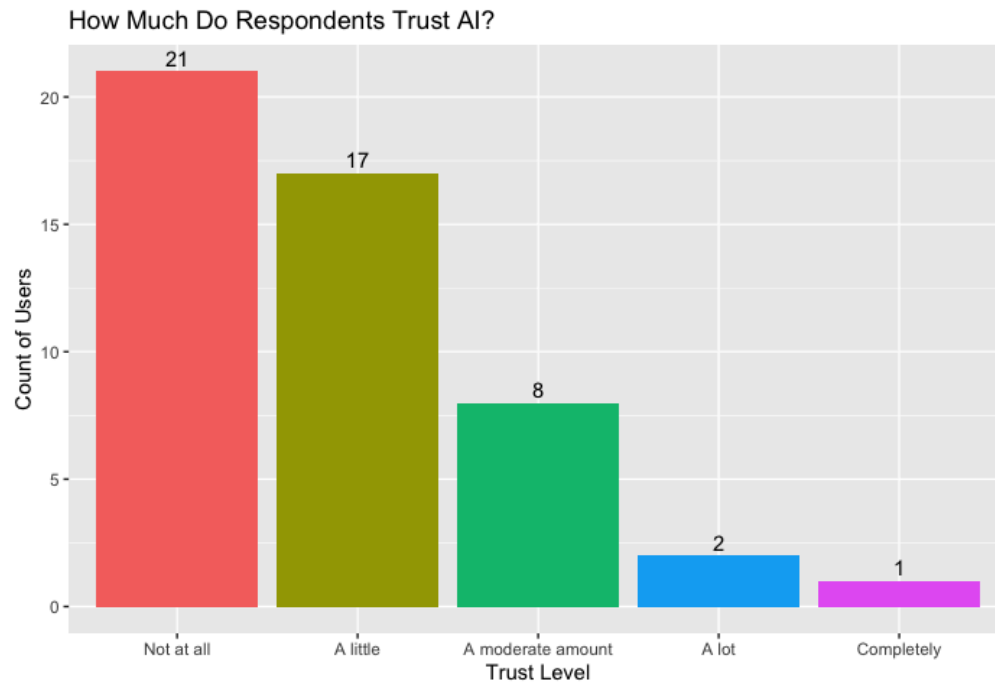


Fig. 4: Distribution of Respondents' Trust in AI

To further our investigation, we then tested whether AI usage frequency was associated with trust in AI to see if frequent AI usage affected users' risk perception.

Fig. 5 shows that the majority of users, despite their AI usage frequency, claimed not to trust AI. The small number of respondents who reported high trust in AI was concentrated among those who reported more frequent AI use.

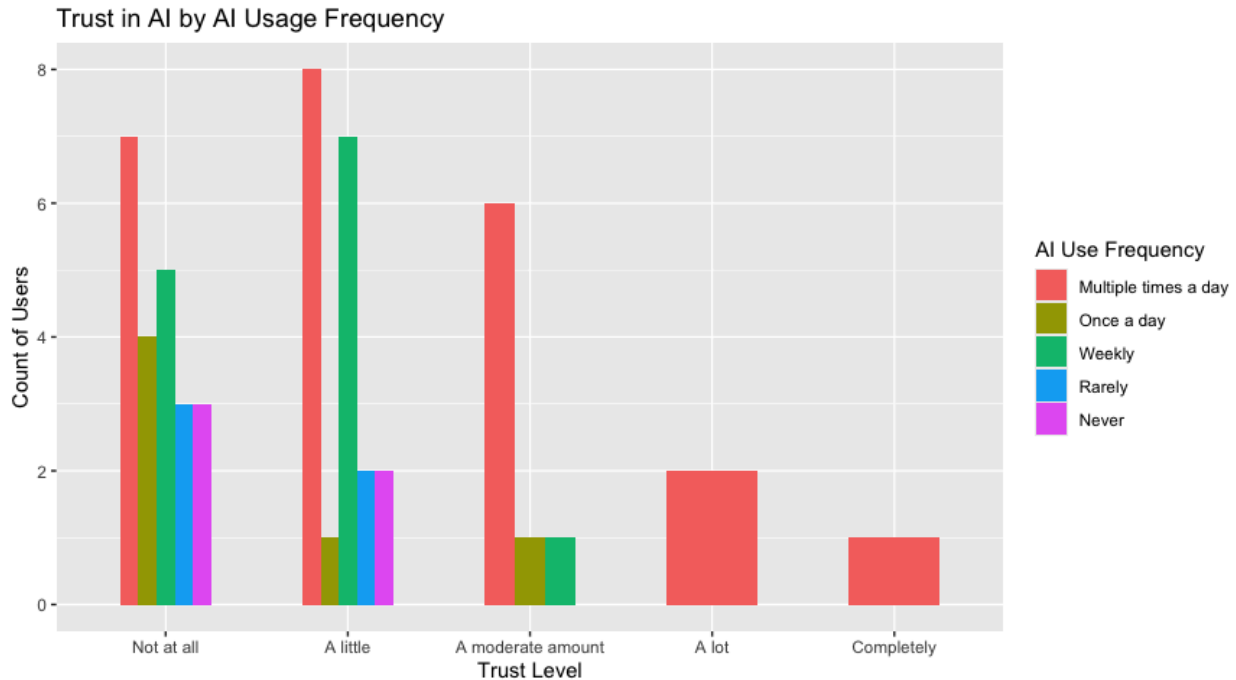


Fig. 5: Distribution of respondents' trust in AI by their AI usage frequency.

To determine whether the AI usage frequency was associated with users' trust in AI from Fig 5 showed any statistical association, a Chi-Square test was conducted.

Chi-Square Test: AI Usage Frequency vs. Trust in AI

statistic	df	p_value
11.5615	16	0.7736

Table II: Chi-square test results for AI usage frequency and trust in AI.

Using the Chi-Square test, we did not find a statistically significant association between AI usage frequency and trust in AI in this sample, as the P-value was .8084, which is well above .05. This suggests that users who reported using AI more often still feel as though AI has risks.

Overall, the data suggest that users are aware that using AI carries risks, and frequent AI usage does not alter whether their risk perception fluctuates.

C. RQ3: "If Aware, Do They Care?" (Awareness vs. Behavior)

In this subsection, we explore whether users' distrust of AI affects their data input behavior.

In Table II, we concluded that frequent AI usage does not affect whether they trust AI. Users who frequently use AI are not more likely to distrust AI than those who do not frequently use it. However, despite this, the frequent usage may lead them to input sensitive documentation.

The respondents of our survey were asked to check off the following data they have shared with an AI: passwords/credentials, personal documents (resumes, IDs, etc.), work/school information, medical/legal information, or none. Asking such questions allowed us to classify which forms of data are the most common for users to leak.

Summary of Trust and Sensitive Information Sharing Among AI Users

Finding	Percent
Frequent AI users with low trust in AI	62.5%
Frequent AI users who input personal documents into AI	55%
Frequent AI users who input at least one sensitive information type into AI	63%
Once-a-day users who input at least one sensitive information type into AI	14%

Table III: Summary of users' behaviors when inputting sensitive information in AI.

Despite 62.5% of frequent AI users from Table III claiming they do not trust AI much, 55% of these frequent users also claimed to have input personal documents at least once to AI.

We also found that 63% of frequent AI users have input at least one form of sensitive documentation into AI, compared to only 14% of once-a-day users performing the same behaviors.

The data suggests that the AI usage of a user may affect whether they input certain forms of personal information. To strengthen the assumption, Table IV summarizes a Chi-Square test, testing whether there is a real association between the factors.

Chi-Square Tests: AI Usage Frequency vs. Q11 Sharing Categories

category	statistic	df	p_value
Passwords or credentials	1.2801	4	0.8647
Personal documents	14.0350	4	0.0072
Work/school confidential info	7.1766	4	0.1268
Medical/legal info	6.2945	4	0.1782
None	14.8243	4	0.0051

Table IV: Chi-Square test results for AI usage frequency and sharing data categories.

The chi-square analysis showed that the personal information and none categories were significantly associated with AI usage frequency, while the other categories were not. This suggests that the frequency of AI use may be related to some forms of information-sharing behavior.

We also compared reported trust in AI with the types of information users reported sharing to analyze whether users' risk perception affects specific forms of information they input into AI. To do so, we conducted a Chi-Square test, summarized in Table V, to determine whether there is an association.

Chi-Square Tests: Trust in AI vs. Q11 Sharing Categories

category	statistic	df	p_value
Passwords or credentials	18.2967	4	0.0011
Personal documents	15.6395	4	0.0035
Work/school confidential info	11.9900	4	0.0174
Medical/legal info	13.3702	4	0.0096
None	14.0300	4	0.0072

Table V: Chi-Square test results for AI usage frequency and trust levels amongst respondents.

Using the Chi-Square test, we can see that all of the input categories are statistically significantly correlated with trust in AI. The p-values for all categories are significantly below .05, indicating that the user's trust level was linked to whether they entered this information into AI.

In short, users' awareness of the risks of AI does indeed affect whether they input sensitive information into AI. Additionally, we found that users who frequently use AI are more likely to input personal documents than those who use AI less often.

D. RQ4: "Who Is Most Susceptible?" (Demographic Analysis)

We next analyzed the respondents' responses to knowledge questions related to AI data storage, data usage, and accuracy. These questions were used to identify possible misconceptions about how AI systems work.

We first determined which age range (Fig. 6) made up the proportion of respondents who incorrectly answered the knowledge questions.

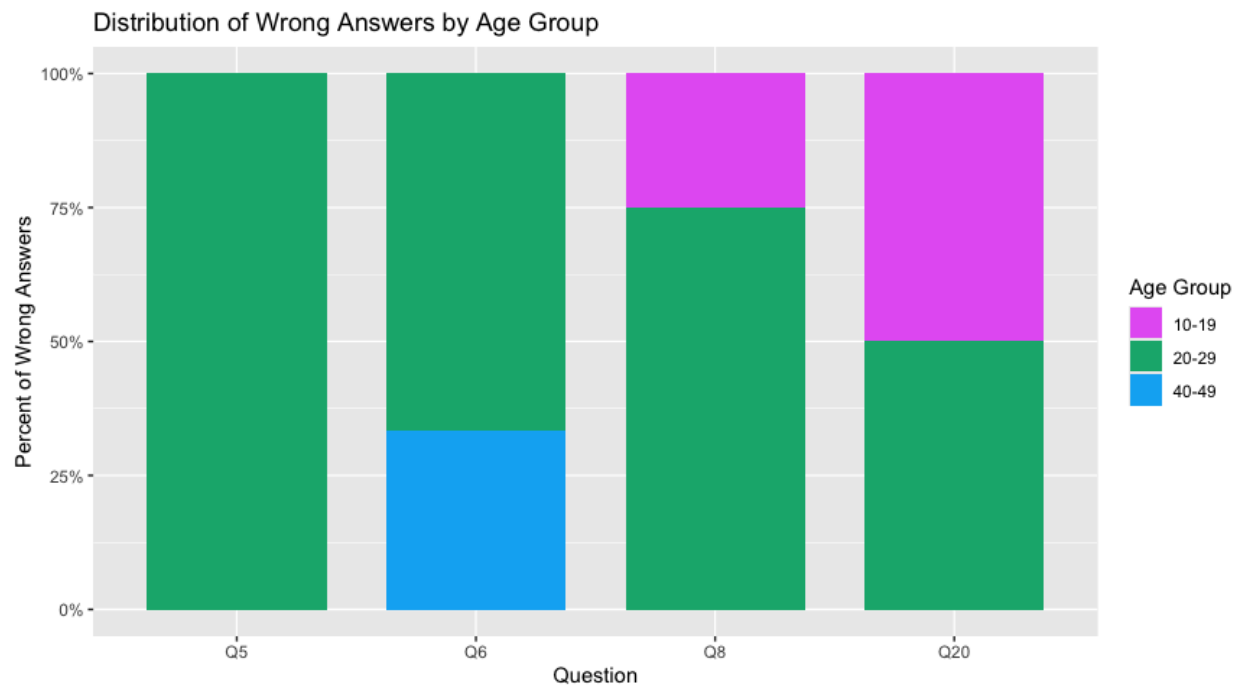


Fig. 6: Proportional distribution of respondents who answered incorrectly by age group.

We then determined which education level (Fig. 7) made up the proportion of respondents who incorrectly answered the knowledge questions.

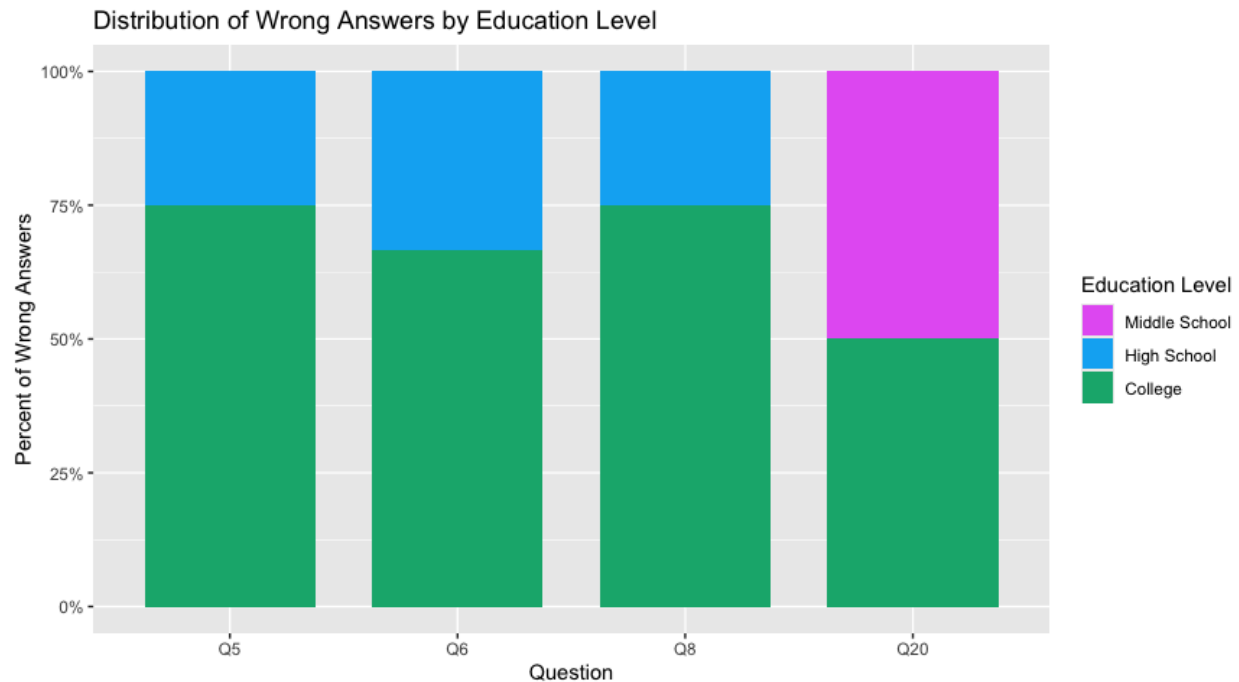


Fig. 7: Proportional distribution of respondents who answered incorrectly by education level.

The distribution of wrong answers was concentrated among respondents aged 20–29 and among respondents with college-level education. However, because these groups also made up a large share of the sample, this alone does not prove that they were inherently more susceptible from this sample.

To strengthen our investigation, we revisited the misconception scores from Table I to see which age group and education level populated the results. The results are visualized in Fig. 8 for the education level.

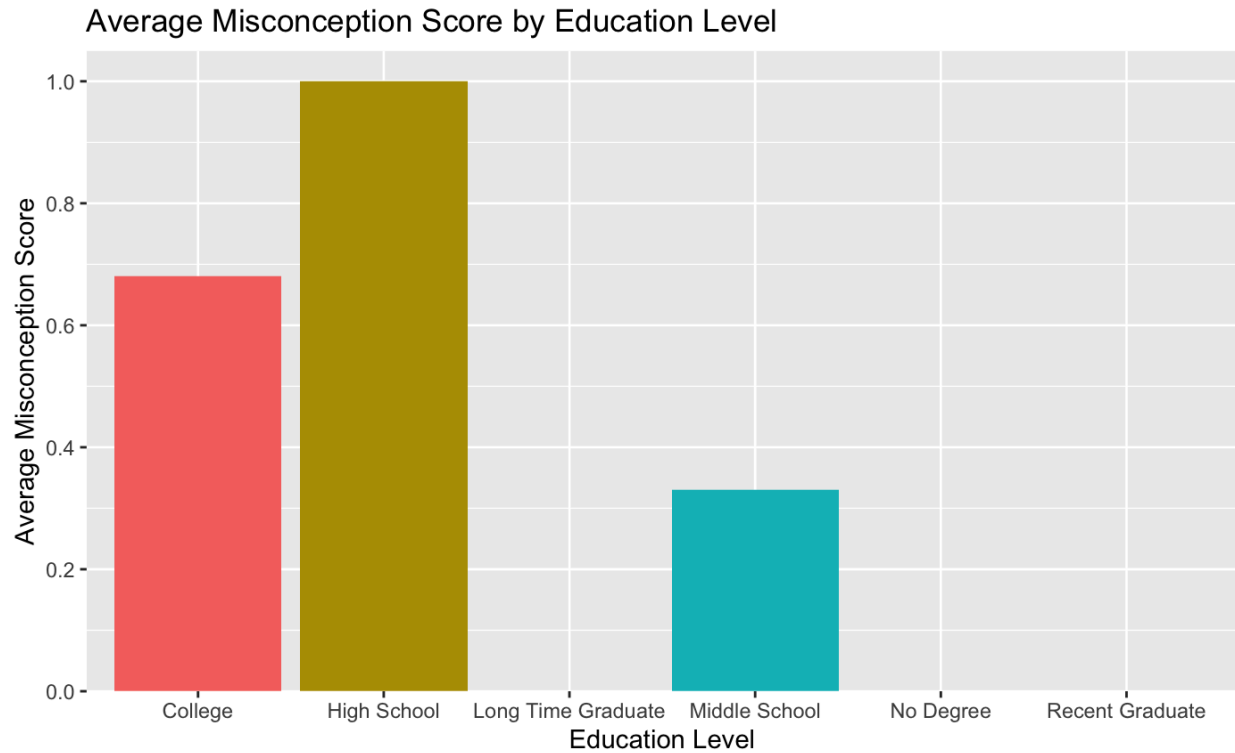


Fig. 8: Distribution of average misconception scores sorted by education level.

From this, we found that high school students and college students had the highest level of misconceptions compared to other education levels. Our respondents mostly consisted of high school and college students, with fewer long-time graduates, no degree, and recent graduates. Because of this, the data is naturally less populated towards these specific demographics.

Similarly, we compared the misconception scores to the age groups, as shown in Fig. 9.

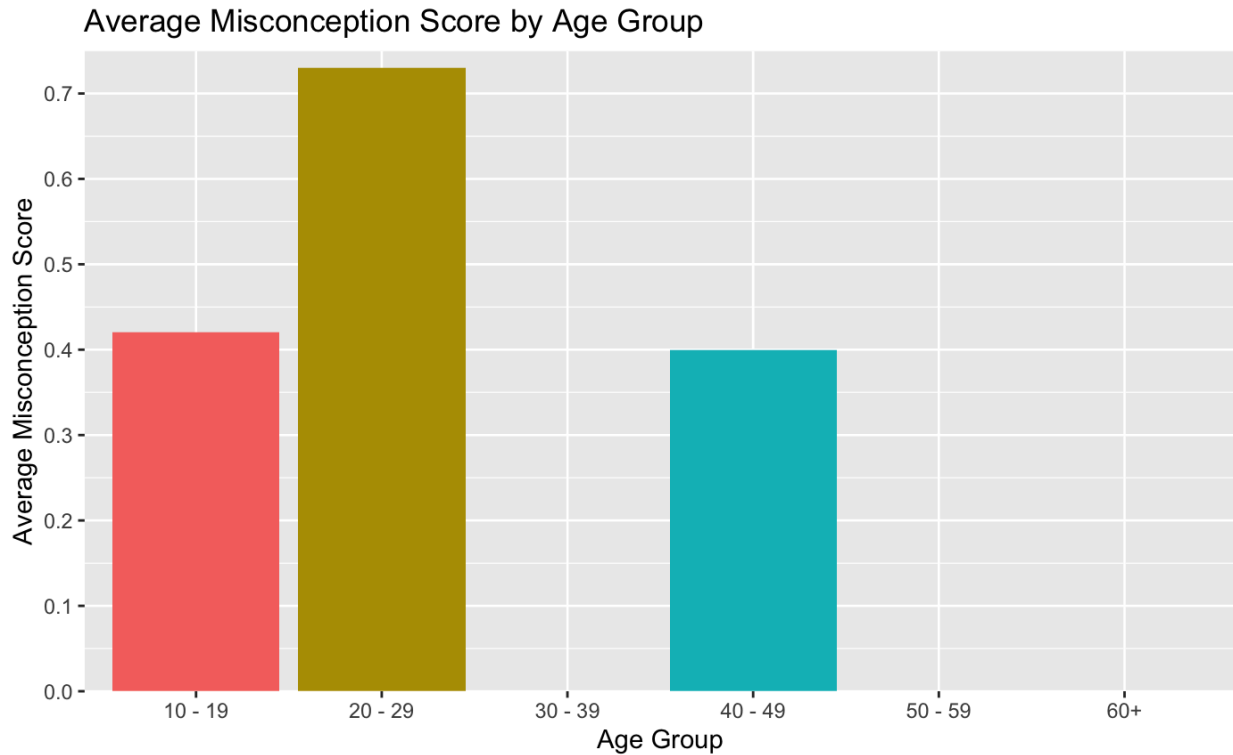


Fig. 9: Distribution of average misconception scores sorted by age groups.

Like Fig. 8, the data for Fig. 9 consisted mostly of those between the ages of 20 and 29. Hence, naturally, that age range had the highest misconception score. Notably, age groups 10 to 19 and 40 to 49 had the same misconception score, indicating they have a similar amount of misconceptions on AI data usage.

Overall, from the figures above, we can conclude that high school and college students, typically between the ages of 20 and 29, are the most susceptible to having misconceptions of AI.

E. RQ5: "Does Education Help?" (Intervention Impact)

Finally, we tested whether a short educational intervention changed how likely users were to verify AI-generated text and links before acting on them, as shown in Fig. 10.

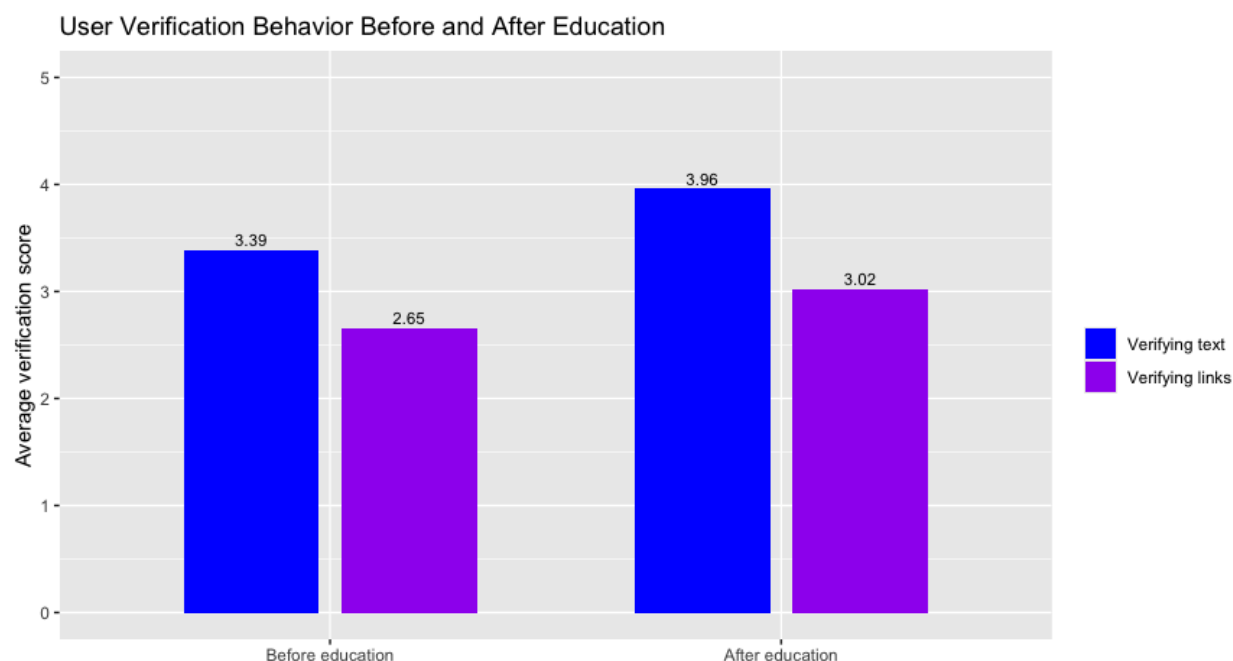


Fig. 10: Visualization of user verification behavior before and after education.

Visually, the appearance of the post-education bars for both verifying texts and verifying links is slightly higher. While analyzing only the averages, which are placed above the bars, we can see 3.39 shoot up to 3.96 and 2.65 shoot up to 3.02, post-education.

However, to determine whether this change was statistically significant, we conducted a paired nonparametric test. Because these are paired before-and-after responses from the same participants, a paired Wilcoxon signed-rank test was used, as summarized in Table VI.

Paired Wilcoxon Tests: Before vs. After Education

measure	statistic	p_value
Verifying text	81.0	0.0157
Verifying links	37.5	0.0003

Table VI: Wilcoxon test results for respondents' answers to verifying text and links from AI before and after being educated.

The educational intervention significantly improved reported verification behavior for both text and links. The change was statistically significant for text and also statistically significant for links, with the link result showing especially strong evidence of

improvement. These findings suggest that even a brief educational prompt may improve safer AI-use behavior.
